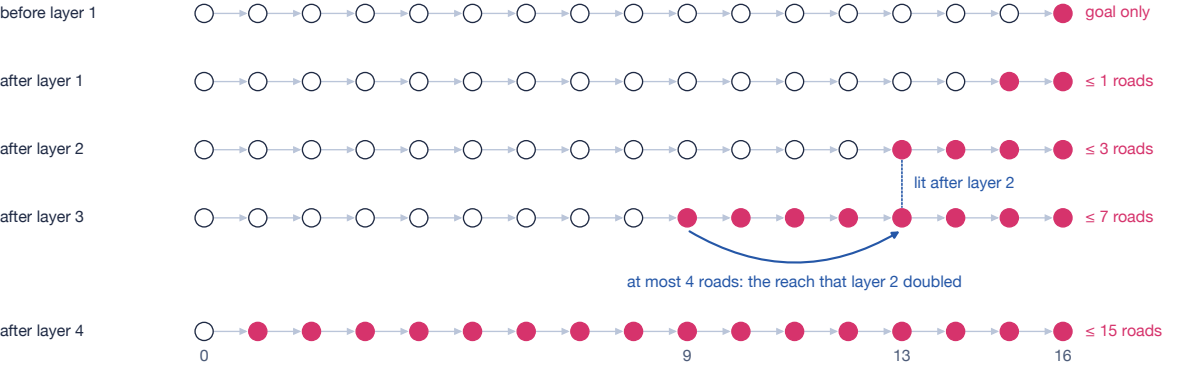
Each layer doubles how far the goal can be seen (a path of 17 towns, goal at town 16)



Town 9 is lit after layer 3: it reaches town 13 within 4 roads, and town 13 was within 3 roads of the goal after layer 2; $4 + 3 = 7 = 2^3 - 1$.